Arjun P

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Education

Mar Baselios College of Engineering and Technology Nalanchira, Trivandrum, Kerala,India
BTech. Computer Science and Engineering with Artificial Intelligence *2022 – 2026*

Experience

Software Development Engineer Oct 2025 – Present
Sacom Technologies *Onsite*

- Working on maintaining and improving production web applications used by active users.
- Fixing bugs, optimizing existing features, and assisting in stabilizing the production codebase.
- Collaborating with the development team using Git workflows including branching, rebasing, and resolving merge conflicts.
- Contributing to frontend improvements using modern web technologies and debugging issues reported by users.

AI Trainee – Intel® Industrial Training Program May 2025 – June 2025
Intel® Unnati (Remote)

- Completed Intel® Unnati training program with focus on applied AI concepts and hands-on project work.
- Built a learning assistant chatbot with voice input, image-based responses, and natural language interaction using LLaMA models via OpenRouter API.
- Implemented an engagement detection system using OpenCV and a Facial Expression Recognition (FER) model to analyze user attentiveness.
- Integrated speech-to-text, text-to-speech, and Streamlit UI for interactive real-time communication.

Web Developer Intern March 2025 – May 2025
Obsidyne Co. (Remote) *Trivandrum, Kerala*

- Familiarity with frontend frameworks like React.js and Next.JS
- Experience in converting Figma designs into fully functional web pages.
- Utilized Tailwind CSS's utility-first approach to rapidly style components with minimal custom CSS.
- Built fully responsive UIs using Tailwind's mobile-first breakpoints and grid/flex utilities.

CyberSecurity Intern December 2024
ICT Academy of Kerala *Trivandrum, Kerala, India*

- Conducted detailed network reconnaissance using Nmap to identify open ports, running services, and OS fingerprints across internal and external networks.
- Analyzed scan results to detect common vulnerabilities such as outdated services, exposed ports, and weak configurations.
- Used Metasploit for exploitation of known vulnerabilities (e.g., MS08-067, EternalBlue) in lab environments to simulate real-world attack vectors.
- Performed end-to-end exploitation including scanning, enumeration, payload generation, and gaining shell access on target systems.

Projects

MUSICED – Social Network for Music | *Next.js 16, TypeScript, Supabase, PostgreSQL, Tailwind*

- Built and deployed a full-stack social platform for music: listening diary, half-star ratings, reviews with threaded comments, ranked lists, follows and an activity feed.
- Integrated the MusicBrainz and Cover Art Archive APIs with a rate-limited provider layer and idempotent, transactional album import keyed by external ID.
- Secured every table with PostgreSQL row-level security and column-level grants; database triggers maintain rating stats, counters, feeds and notifications.
- Implemented live notifications with Supabase Realtime, Bayesian-ranked charts, taste-based recommendations, a moderation console and Postgres-backed rate limiting.
- Live Site | Repo Link

College Website Development (at Obsidyne) | *ReactJS, TailwindCSS, Typescript*

- Designed a fully responsive College website using ReactJS, Tailwind & Typescript to ensure optimal viewing across all devices
- Converted Figma wireframes into production-ready pages using React.js, Next.js, and Tailwind CSS.
- Implemented mobile-first layouts with Tailwind's grid/flex utilities for seamless user experience across devices.

Healthcare Appointment Management System | *ReactJS, Django, Postgres*

- Built a full-stack platform enabling secure doctor–patient appointment scheduling with JWT authentication and role-based access control.
- Implemented real-time conflict detection, doctor availability management, and appointment status workflows using Django REST Framework.
- Designed responsive React dashboards for patients and doctors with seamless booking, profile management, and real-time updates.
- Utilized PostgreSQL for efficient data handling and integrated RESTful APIs for reliable communication between frontend and backend.
- Repo Link

Kudukka – AI-Powered Crypto Investment and Staking | *Python, thirdweb, ReactJS*

- Developed an LSTM-based model to forecast future cryptocurrency prices (e.g., BTC, ETH) using historical time series data.
- Collected real-time tweets using Twitter API and performed sentiment analysis using a hybrid model combining LSTM and VADER (for text context) .
- Enhanced sentiment prediction accuracy by fusing deep contextual understanding (LSTM) with rule-based scoring (VADER) to detect market mood shifts.
- Built a decision logic layer to recommend the most promising cryptocurrency based on predicted price trends and public sentiment alignment.
- Working Demo

EduLlama - AI Powered Learning Assistant | *Python,OpenCV, LLaMA (OpenRouter),FER* May 2025

- Developed a learning assistant chatbot that supports voice input, image-based responses, and natural language interaction using LLaMA models via OpenRouter API.
- Implemented an engagement detection system using OpenCV and a Facial Expression Recognition (FER) model to classify user engagement from emotional cues.
- Integrated Python libraries such as SpeechRecognition for speech-to-text, gTTS for text-to-speech, and Streamlit for building an interactive UI.
- Designed a workflow for real-time adaptation, where engagement feedback influences the chatbot's responses to maintain student attention.
- Optimized performance and responsiveness of the assistant by managing API calls, lightweight preprocessing, and efficient use of Python libraries.

AI-Powered Swarm Robot for Search and Rescue | *ROS2, Gazebo, TensorFlow Lite, YOLO, Raspberry Pi, MQTT*

- Developed an AI-powered multi-robot swarm system for autonomous search and rescue, enabling decentralized coordination and real-time decision making.
- Implemented YOLO-based object detection with TensorFlow Lite and multi-sensor fusion (ultrasonic, IR, camera) for victim detection, achieving over 93% detection accuracy.
- Applied Reinforcement Learning and the A* algorithm for adaptive path planning and efficient area exploration, achieving over 90% coverage while reducing redundant exploration by 30%.
- Simulated the system using ROS2 and Gazebo and deployed it on Raspberry Pi-based robots with MQTT-based wireless communication for real-time multi-robot coordination.

Know-Note - A simple Note Web App | *NextJS,TypeScript,MongoDB*

- Built a full-stack note-taking web app using Next.js, TypeScript, and MongoDB with responsive UI.
- Implemented CRUD features with clean, modular architecture.
- Ensured a clean and organized codebase for easy maintenance.

Personal Portfolio Website | *ReactJS, TailwindCSS, Typescript*

- Designed a fully responsive website using ReactJS, Tailwind & Typescript to ensure optimal viewing across all devices

Mileage Prediction using Different ML Models | *Python, Machine Learning*

- Predicted Mileage of the cars using different Machine Learning Algorithms
- Repo Link

Technical Skills

Languages: Python, C, SQL (MYSQL, Postgres), JavaScript, HTML/CSS
Frameworks: React, Node.js, ElectronJS, TailwindCSS, MetaSploit, Django
Developer Tools: Git, Linux, Bash, VS Code, PyCharm
Machine Learning & AI: pandas, PyTorch, NumPy, Matplotlib, Tensorflow, Sklearn, LLMs , Prompt Engineering